



MATHEW KIPLAGAT KIMUTAI

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CAREER PROFILE AND SUMMARY

As a qualified industrial chemist with extensive experience in laboratory operations, process optimization, and chemical manufacturing, I deliver precise and efficient results in fast-paced environments. My core competencies include analytical techniques such as HPLC, GC, and titration; quality control and assurance; process development and scale-up; formulation optimization; regulatory compliance with GMP and EH&S standards; and data analysis with technical reporting. Key achievements involve developing and optimizing chemical processes to enhance yield, efficiency, and product quality; implementing robust quality controls while ensuring adherence to safety and environmental regulations; conducting quantitative and qualitative analyses, method validation, and troubleshooting; streamlining laboratory and production workflows; and providing technical support to cross-functional teams in research and development initiatives for new products and materials. I am seeking a challenging role in industrial chemistry where I can apply innovative solutions and best practices to drive operational excellence, sustainability, and organizational growth.

EDUCATIONAL BACKGROUND

May 2021 to 2025	BSc Industrial Chemistry Dedan Kimathi University of Technology (Second class owners)
2014 to 2018	Kenya certificate of Secondary Education, KCSE (Overall Grade: B-) Kipsoen Boys High School
2005 to 2014	Kenya Certificate of Primary Education, KCPE (Overall score:329 marks out of 500) Kapsoiyo Primary School

KEY SKILLS AND COMPETENCIES ACQUIRED

Analytical Chemistry: Proficient in advanced analytical techniques including HPLC, GC, spectroscopy, and titration; experienced in method development, validation, and troubleshooting to ensure accurate qualitative and quantitative results.

Process Optimization and Scale-Up: Skilled in designing, optimizing, and scaling chemical processes to improve yield, efficiency, sustainability, and product quality in industrial manufacturing environments.

Quality Control and Regulatory Compliance: Expertise in implementing GMP, ISO standards, and EH&S regulations; experienced in quality assurance testing, batch release, and documentation for regulatory submissions.

Research and Development (R&D): Hands-on experience using RDKit for cheminformatics tasks, including molecular modelling, virtual screening, descriptor calculation, and structure-based drug/material design.

Detail-Oriented and Analytical Thinking: Strong ability to analyse complex problems, assess risks, and derive data-driven solutions with meticulous attention to accuracy in experimental and computational work.

Data Analytics with Python: Advanced proficiency in Python for scientific data analysis using libraries such as Pandas, NumPy, SciPy, and Matplotlib; skilled in processing large datasets, statistical analysis, and generating insightful visualizations.

Machine Learning Development: Experienced in building and deploying machine learning models with Python (scikit-learn, TensorFlow) for predictive modelling, classification, and regression applied to chemical and process data.

ICT Competence: Proficient in Microsoft Office Suite, scientific software, version control (Git), cloud platforms, and collaborative tools; adept at integrating IT solutions to enhance laboratory and R&D workflows.

WORK HISTORY

PROCESS ENGINEER

BIOCORN PRODUCTS EPZ LIMITED OCTOBER 2023 TO DECEMBER 19 2023

Duties and Responsibilities

Cob handling: quality checks, offloading, conveying, storage systems, stockpile management and safety processes

Cob processing: cob transfer systems, quality tests at cob stockpiles, size reduction operations, milling technology silos and material conveying systems.

Hydrolysis: Hydrolysis chemistry, conversion of pentosane to pentose and to furfural and other carboxylic acids, reaction kinetics, reaction parameters, reactor types and capacity, mode of operations Filiation systems, cooling and heat recoveries, condensation, quality checks and analysis

Steam generation: Steam quality, pressures, temperatures and safe practices

Distillation: Dynamics and control, distillation columns, condensers, reboilers, decanters, reflux ratios, tanks, vessels, safety practices and quality checks performed

Fluid Transfer Systems: Pumps, pipeline codes, valves, safety valves and their functions, transfer rates and safety practices

Effluent treatment plant

1. All steps involved in the effluent treatment; screening, agitation, coagulation, flocculation, clarification, aeration and filtration
2. Biological oxygen demand (BOD) and chemical oxygen demand (COD)procedures, processes and tests

Quality control

1. Use of gas chromatography machine to determine purity of the tested product
 2. Use of the Karl Fischer machine to determine the moisture content of furfural
- Water analysis test done using the Palin kit

PRODUCTION ENGINEER

EMMAT AGROFARM LIMITED AUGUST 2025 TO NOVEMBER 2025

Quality Assurance and Compliance — Ensure final products meet specifications and regulatory standards (e.g., nutrient content, granule size, purity). Monitor quality throughout the process, conduct or oversee testing, and address any non-conformances to prevent rejects or recalls.

REFEREES

1. DR. SAMUEL MIRIE,

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2. DR. PAUL TANUI, Senior Lecturer Dedan Kimathi University,

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